SLOT – L33+L34

VIT
Vellore Institute of Technology
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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT – I – FALL SEMESTER 2026-2027

Programme Name & Branch: M.Tech. (Integrated) CSE (Data Science)

Course Name: Data Science Programming Lab

Course Code: CSI3004

Assessment 1: House Rent Prediction Using R (Linear Regression)

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Data Understanding

data = read.csv("/home/matlab/Downloads/House_Rent_Dataset.csv")

#Structure of data
str(data)
#Summary of data
summary(data)

OUTPUT :

```
Console   Terminal ×   Background Jobs ×
R 4.6.1 · ~/
> data = read.csv("/home/matlab/Downloads/House_Rent_Dataset.csv")
>
> #Structure of data
> str(data)
'data.frame':   4746 obs. of  12 variables:
 $ Posted.On        : chr  "2022-05-18" "2022-05-13" "2022-05-16" "2022-07-04" ...
 $ BHK              : int  2 2 2 2 2 2 2 1 2 2 ...
 $ Rent             : int  10000 20000 17000 10000 7500 7000 10000 5000 26000 10000 ...
 $ Size             : int  1100 800 1000 800 850 600 700 250 800 1000 ...
 $ Floor            : chr  "Ground out of 2" "1 out of 3" "1 out of 3" "1 out of 2" ...
 $ Area.Type        : chr  "Super Area" "Super Area" "Super Area" "Super Area" ...
 $ Area.Locality    : chr  "Bandel" "Phool Bagan, Kankurgachi" "Salt Lake City Sector 2" "Dumdum Park" ...
 $ City             : chr  "Kolkata" "Kolkata" "Kolkata" "Kolkata" ...
 $ Furnishing.Status: chr  "Unfurnished" "Semi-Furnished" "Semi-Furnished" "Unfurnished" ...
 $ Tenant.Preferred : chr  "Bachelors/Family" "Bachelors/Family" "Bachelors/Family" "Bachelors/Family" ...
 $ Bathroom         : int  2 1 1 1 1 2 2 1 2 2 ...
 $ Point.of.Contact : chr  "Contact Owner" "Contact Owner" "Contact Owner" "Contact Owner" ...
```

```
> #Summary of data
> summary(data)
    Posted.On            BHK              Rent              Size            Floor           Area.Type       Area.Locality
 Length   :4746   Min.   :1.000   Min.   :   1200   Min.   :  10.0   Length   :4746   Length   :4746   Length   :4746
 N.unique :  81   1st Qu.:2.000   1st Qu.:  10000   1st Qu.: 550.0   N.unique : 480   N.unique :   3   N.unique :2235
 N.blank  :   0   Median :2.000   Median :  16000   Median : 850.0   N.blank  :   0   N.blank  :   0   N.blank  :   0
 Min.nchar:  10   Mean   :2.084   Mean   :  34994   Mean   : 967.5   Min.nchar:   1   Min.nchar:  10   Min.nchar:   3
 Max.nchar:  10   3rd Qu.:3.000   3rd Qu.:  33000   3rd Qu.:1200.0   Max.nchar:  24   Max.nchar:  11   Max.nchar:  68
                  Max.   :6.000   Max.   :3500000   Max.   :8000.0
      City         Furnishing.Status  Tenant.Preferred     Bathroom        Point.of.Contact
 Length   :4746   Length   :4746    Length   :4746    Min.   : 1.000   Length   :4746
 N.unique :   6   N.unique :   3    N.unique :   3    1st Qu.: 1.000   N.unique :   3
 N.blank  :   0   N.blank  :   0    N.blank  :   0    Median : 2.000   N.blank  :   0
 Min.nchar:   5   Min.nchar:   9    Min.nchar:   6    Mean   : 1.966   Min.nchar:  13
 Max.nchar:   9   Max.nchar:  14    Max.nchar:  16    3rd Qu.: 2.000   Max.nchar:  15
                                                      Max.   :10.000
```

Data Preprocessing :

data_clean = subset(data,select = -c(Posted.On, Area.Locality, Floor))

data_clean$Area.Type = as.factor(data_clean$Area.Type)
data_clean$City = as.factor(data_clean$City)
data_clean$Furnishing.Status = as.factor(data_clean$Furnishing.Status)
data_clean$Tenant.Preferred = as.factor(data_clean$Tenant.Preferred)
data_clean$Point.of.Contact = as.factor(data_clean$Point.of.Contact)

data_clean = na.omit(data_clean)

set.seed(123)
sample_size <- floor(0.7 * nrow(data_clean))
train_indices <- sample(seq_len(nrow(data_clean)), size = sample_size)

train_data <- data_clean[train_indices,]
test_data <- data_clean[-train_indices,]

cat("Training data rows:", nrow(train_data), "\n")
cat("Testing data rows:", nrow(test_data), "\n")

OUTPUT:

```
> data_clean = subset(data,select = -c(Posted.On, Area.Locality, Floor))
> 
> data_clean$Area.Type = as.factor(data_clean$Area.Type)
> data_clean$City = as.factor(data_clean$City)
> data_clean$Furnishing.Status = as.factor(data_clean$Furnishing.Status)
> data_clean$Tenant.Preferred = as.factor(data_clean$Tenant.Preferred)
> data_clean$Point.of.Contact = as.factor(data_clean$Point.of.Contact)
> 
> data_clean = na.omit(data_clean)
> 
> set.seed(123)
> sample_size <- floor(0.7 * nrow(data_clean))
> train_indices <- sample(seq_len(nrow(data_clean)), size = sample_size)
> 
> train_data <- data_clean[train_indices, ]
> test_data <- data_clean[-train_indices, ]
> 
> cat("Training data rows:", nrow(train_data), "\n")
Training data rows: 3322
> cat("Testing data rows:", nrow(test_data), "\n")
Testing data rows: 1424
> 
```

Model Building :

lm_model = lm(Rent ~ ., data = train_data)

summary(lm_model)

OUTPUT:

```
> lm_model = lm(Rent ~ ., data = train_data)
>
> summary(lm_model)

Call:
lm(formula = Rent ~ ., data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-173656  -20601   -2569   12066 3385303

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)
(Intercept)                     -13732.476  51941.116  -0.264 0.791499
BHK                               2903.189   2723.673   1.066 0.286542
Size                                41.005      3.263  12.566  < 2e-16 ***
Area.TypeCarpet Area             -7926.883  51609.698  -0.154 0.877940
Area.TypeSuper Area             -11822.466  51544.527  -0.229 0.818600
CityChennai                      -8318.704   4188.183  -1.986 0.047090 *
CityDelhi                         7755.104   4777.068   1.623 0.104599
CityHyderabad                   -16943.920   4268.212  -3.970 7.35e-05 ***
CityKolkata                      -3407.306   5046.198  -0.675 0.499582
CityMumbai                       49014.919   4786.265  10.241  < 2e-16 ***
Furnishing.StatusSemi-Furnished  -5464.483   3834.035  -1.425 0.154177
Furnishing.StatusUnfurnished     -6479.261   3963.485  -1.635 0.102199
Tenant.PreferredBachelors/Family   272.336   3611.052   0.075 0.939887
Tenant.PreferredFamily           -8319.896   5129.718  -1.622 0.104920
Bathroom                         10199.426   2825.385   3.610 0.000311 ***
Point.of.ContactContact Owner   -10576.642   3866.126  -2.736 0.006258 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 72780 on 3306 degrees of freedom
Multiple R-squared:  0.2743,	Adjusted R-squared:  0.271
F-statistic: 83.31 on 15 and 3306 DF,  p-value: < 2.2e-16

>
```

Prediction & Evaluation

test_data <- subset(test_data, Point.of.Contact != "Contact Builder")

predictions = predict(lm_model , newdata = test_data)
actuals = test_data$Rent

rmse = sqrt(mean((predictions - actuals)^2, na.rm = TRUE))

mae = mean(abs(predictions - actuals), na.rm = TRUE)

rss = sum((predictions - actuals)^2 , na.rm = TRUE)
tss = sum((actuals - mean(actuals))^2, na.rm = TRUE)
r_squared = 1 - (rss/tss)

```
cat("\n--- Model Evaluation ---\n")
cat("RMSE:", rmse, "\n")
cat("MAE:", mae, "\n")
cat("R-squared:", r_squared, "\n")
```

OUTPUT :

```
> test_data <- subset(test_data, Point.of.Contact != "Contact Builder")
>
> predictions = predict(lm_model , newdata = test_data)
> actuals = test_data$Rent
>
> rmse = sqrt(mean((predictions - actuals)^2, na.rm = TRUE))
>
> mae = mean(abs(predictions - actuals), na.rm = TRUE)
>
> rss = sum((predictions - actuals)^2 , na.rm = TRUE)
> tss = sum((actuals - mean(actuals))^2, na.rm = TRUE)
> r_squared = 1 - (rss/tss)
>
> cat("\n--- Model Evaluation ---\n")

--- Model Evaluation ---
> cat("RMSE:", rmse, "\n")
RMSE: 39971.88
> cat("MAE:", mae, "\n")
MAE: 22298.14
> cat("R-squared:", r_squared, "\n")
R-squared: 0.5277052
>
```

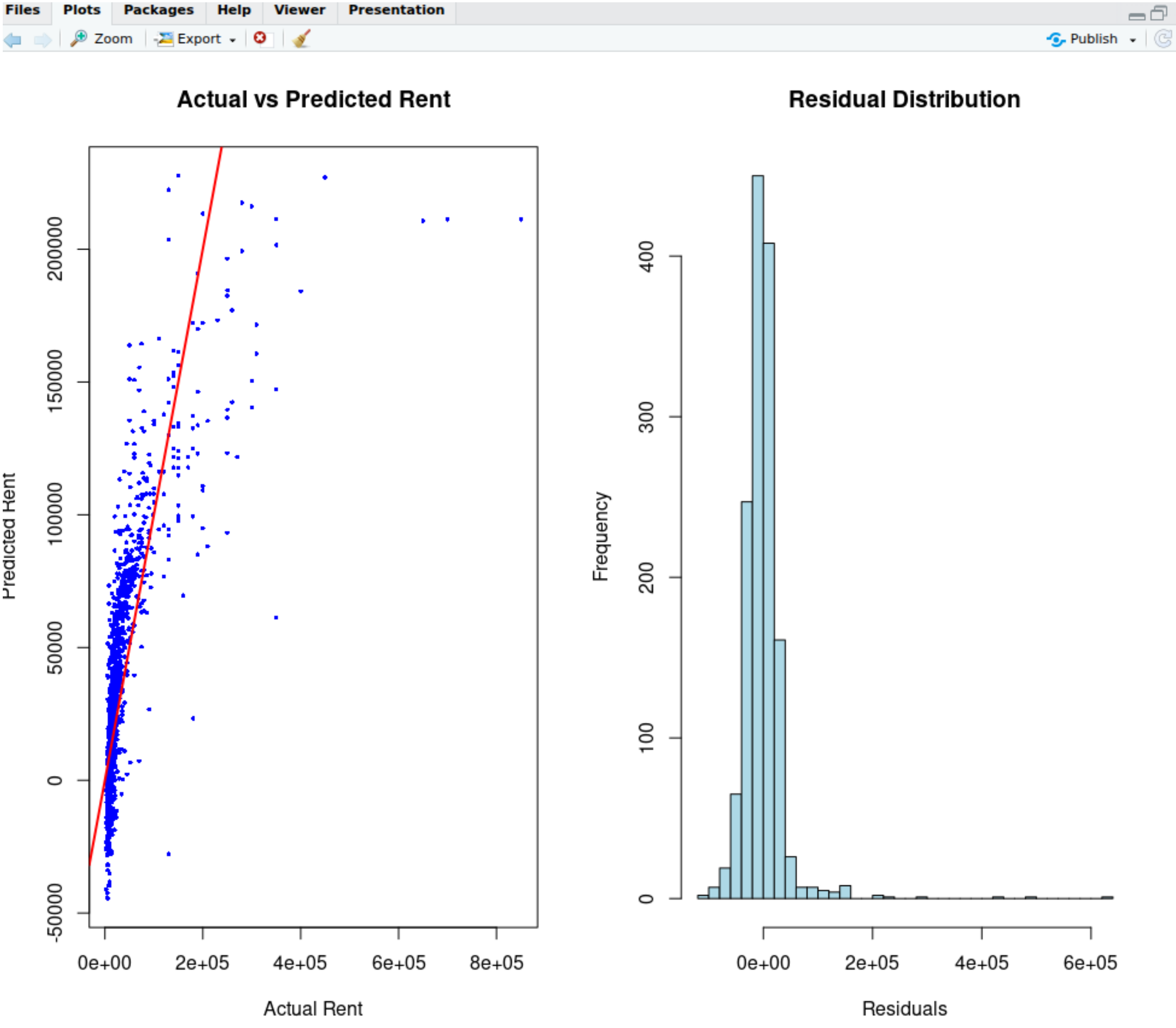
Visualization

```
# ------------------------------------------
# Step 5: Visualization
# ------------------------------------------
# Set up the plotting area to show 2 plots side-by-side
par(mfrow = c(1, 2))

# 1. Actual vs Predicted values
plot(actuals, predictions,
     main = "Actual vs Predicted Rent",
     xlab = "Actual Rent",
     ylab = "Predicted Rent",
     col = "blue", pch = 16, cex = 0.5)
# Add a red line showing perfect predictions
abline(a = 0, b = 1, col = "red", lwd = 2)

# 2. Residual distribution
residuals <- actuals - predictions
hist(residuals, breaks = 40,
     main = "Residual Distribution",
     xlab = "Residuals",
     col = "lightblue", border = "black")
```

OUTPUT :



Model Improvement

```
# ------------------------------------------
# Step 6: Model Improvement
# ------------------------------------------

# 1. Rebuild the model removing insignificant variables
improved_model <- lm(Rent ~ Size + City + Bathroom + Point.of.Contact, data = train_data)

# Display the summary of the new, simplified model
summary(improved_model)

# 2. Predict on test data using the improved model
predictions_improved <- predict(improved_model, newdata = test_data)

# 3. Compute Evaluation Metrics to compare with the original model
# RMSE
rmse_improved <- sqrt(mean((predictions_improved - actuals)^2, na.rm = TRUE))
```

# MAE
mae_improved <- mean(abs(predictions_improved - actuals), na.rm = TRUE)

# R-squared
rss_improved <- sum((predictions_improved - actuals)^2, na.rm = TRUE)
r_squared_improved <- 1 - (rss_improved / tss)

# 4. Compare Results
cat("\n--- Comparison of Models ---\n")
cat("Original RMSE:", rmse, " | Improved RMSE:", rmse_improved, "\n")
cat("Original MAE: ", mae, " | Improved MAE: ", mae_improved, "\n")
cat("Original R2: ", r_squared, " | Improved R2: ", r_squared_improved, "\n")

OUTPUT :

```
> # 1. Rebuild the model removing insignificant variables
> improved_model <- lm(Rent ~ Size + City + Bathroom + Point.of.Contact, data = train_data)
>
> # Display the summary of the new, simplified model
> summary(improved_model)

Call:
lm(formula = Rent ~ Size + City + Bathroom + Point.of.Contact,
    data = train_data)

Residuals:
    Min      1Q  Median      3Q     Max
-198870  -20230   -2668   12096 3385914

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                 -26712.516   5382.799  -4.963 7.31e-07 ***
Size                            42.050      3.128  13.445  < 2e-16 ***
CityChennai                  -8731.838   4152.027  -2.103 0.035538 *
CityDelhi                     9234.517   4703.595   1.963 0.049696 *
CityHyderabad               -17428.863   4212.427  -4.137 3.60e-05 ***
CityKolkata                  -2012.458   4865.210  -0.414 0.679163
CityMumbai                   49396.103   4670.230  10.577  < 2e-16 ***
Bathroom                     11890.783   2282.660   5.209 2.01e-07 ***
Point.of.ContactContact Owner -12299.433   3448.016  -3.567 0.000366 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 72800 on 3313 degrees of freedom
Multiple R-squared:  0.2723,	Adjusted R-squared:  0.2706
F-statistic:   155 on 8 and 3313 DF,  p-value: < 2.2e-16
```

```
>
> # 2. Predict on test data using the improved model
> predictions_improved <- predict(improved_model, newdata = test_data)
>
> # 3. Compute Evaluation Metrics to compare with the original model
> # RMSE
> rmse_improved <- sqrt(mean((predictions_improved - actuals)^2, na.rm = TRUE))
>
> # MAE
> mae_improved <- mean(abs(predictions_improved - actuals), na.rm = TRUE)
>
> # R-squared
> rss_improved <- sum((predictions_improved - actuals)^2, na.rm = TRUE)
> r_squared_improved <- 1 - (rss_improved / tss)
>
> # 4. Compare Results
> cat("\n--- Comparison of Models ---\n")

--- Comparison of Models ---
> cat("Original RMSE:", rmse, " | Improved RMSE:", rmse_improved, "\n")
Original RMSE: 39971.88  | Improved RMSE: 40219.83
> cat("Original MAE: ", mae, "  | Improved MAE: ", mae_improved, "\n")
Original MAE:  22298.14   | Improved MAE:  22429.5
> cat("Original R2:  ", r_squared, " | Improved R2:  ", r_squared_improved, "\n")
Original R2:   0.5277052  | Improved R2:   0.5218274
>
```

Model Improvement Visualization

```
# -----------------------------------------
# Step 7: Model Comparison Visualization
# -----------------------------------------

# Set up the plotting area to show 3 plots side-by-side
par(mfrow = c(1, 3))

# Define colors for the bars
colors <- c("skyblue", "salmon")
model_names <- c("Original", "Improved")

# 1. Compare R-Squared (Higher is Better)
r2_values <- c(r_squared, r_squared_improved)
barplot(r2_values,
        names.arg = model_names,
        col = colors,
        main = "R-Squared\n(Higher Better)",
        ylab = "R-squared Score",
        ylim = c(0, 0.6)) # Adjusting Y-axis to see the small difference

# 2. Compare RMSE (Lower is Better)
rmse_values <- c(rmse, rmse_improved)
barplot(rmse_values,
        names.arg = model_names,
        col = colors,
```

```
        main = "RMSE\n(Lower Better)",
        ylab = "Root Mean Squared Error",
        ylim = c(0, 45000)) # Adjusting Y-axis to see the small difference

# 3. Compare MAE (Lower is Better)
mae_values <- c(mae, mae_improved)
barplot(mae_values,
        names.arg = model_names,
        col = colors,
        main = "MAE\n(Lower Better)",
        ylab = "Mean Absolute Error",
        ylim = c(0, 25000)) # Adjusting Y-axis to see the small difference

# Reset graphics device back to 1x1 plotting
dev.off()
```

OUTPUT :